

ERIK TAYLOR, PhD

AI Architect & ML Engineer | Agentic, Edge & Clinical AI Systems | Model Evaluation & Risk

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PhD engineer with 10+ years bringing applied AI from research and architecture to shipped products. Building agentic LLM applications and model-evaluation systems; deploying models to resource-constrained edge hardware — from browser-based LLMs to on-sensor vision. Led >\$2M in NIH-funded computational imaging programs across cardiac, neuro-oncology, and traumatic brain injury, translating research into clinically validated systems with cross-functional teams spanning engineering, clinical, regulatory, and legal stakeholders. At Sony, drove end-to-end development of an on-sensor/on-device vision pipeline and orchestrated a high-stakes program recovery that rescued a flagship pre-market product. Evaluated frontier-model reasoning failures for Snorkel AI to inform RL reward signals and alignment strategy; shipped a privacy-first browser-LLM consumer health product from concept to production. Two decades of peer-reviewed research (50+ publications) ground a pragmatic engineering practice built on measurement, reliability, and safe deployment in regulated environments. Pursuing IAPP AIGP certification (Q3 2026).

PROFESSIONAL EXPERIENCE

AI Research Lead & ML Engineer

Oct 2024 – Present

Taylor ML R&D | San Francisco, CA

- **Agentic AI Systems:** Architected apply-daemon (open-source) — an LLM-orchestrated agentic pipeline that ingests roles and triages through a multi-stage cascading-LLM workflow (extract → judge → evaluate), routing scored matches to a Slack human-in-the-loop layer where one reaction triggers deep-research resume synthesis. Cost-optimized via local small-model inference with on-demand escalation to frontier models.
- **Agentic Development Practice:** Build and ship production repositories primarily through agentic coding workflows — Claude Code (CLI and chat) for architecture, implementation, debugging, and iteration. Operate as the human-in-the-loop director of AI-assisted engineering for apply-daemon and the autonomous trading platform.
- **Technical Writing & Knowledge Leadership:** Author of “Digital Mind,” a technical series on frontier AI architectures (Diffusion Transformers, LLM reasoning, alignment, failure modes) that bridges academic research and practical implementation.

Founder & Lead ML Engineer

Sep 2025 – Feb 2026

Napwise.ai | San Francisco, CA

- **On-Device Generative AI Product:** Shipped a consumer-facing, privacy-first infant sleep assistant running entirely in-browser. Architected a tiered inference engine (WebLLM/WebGPU, models down to Qwen 0.6B) with a hybrid RAG layer combining pre-cached JSON retrieval and LLM reasoning to minimize latency across diverse consumer hardware.
- **Edge Deployment & Cost Strategy:** Owned the transition to on-device AI to eliminate cloud inference costs and ensure data residency. Evaluated and selected efficient open-source models and RAG implementations to maintain performance across heterogeneous environments— BYO compute for LLMs.
- **Full-Stack Product Ownership:** Lead engineer from concept through production: product requirements, React frontend, scheduling engine, Cloudflare deployment, analytics, user testing, and roadmap iteration. Translated user insights into technical requirements to refine reliability and practical utility.

Senior Algorithm Engineer

Jul 2022 – Oct 2024

Sony Electronics | San Jose, CA

- **On-Sensor & On-Device Computer Vision:** Led end-to-end development of a production two-stage vision pipeline for phones and VR headsets — a sub-1 MB CNN running on the image sensor that offloads heavier compute to the host device. Owned data curation, multi-GPU distributed training (DDP), INT8 quantization, and latency optimization (~30 FPS), securing a major B2B design win with a Fortune 500 customer.

- **Foundation-Model Roadmapping & AI Architecture:** Drove the technical roadmap for next-generation AI features by benchmarking SOTA vision and language foundation models – including a working prototype of open-vocabulary detection (OWL-ViT, CLIP) for promptable on-device perception – against performance, safety, and deployment constraints. Translated architectural trade-offs into product strategy.
- **Cross-Functional Program Recovery:** Orchestrated a high-stakes response to a sudden data policy change by bridging legal, business, and academic partners to co-develop a compliant dataset. Unblocked a next-generation AI hardware initiative and rescued a multi-month delivery timeline for a flagship pre-market product.
- **Model Validation & Safety Criteria:** Defined evaluation protocols for edge AI products targeting regulated markets, establishing internal standards for performance, bias, and deployment constraints aligned with responsible AI principles.

AI Evaluation Specialist · Contract

Feb 2025 – Aug 2025

Snorkel AI | Remote

- **Frontier Model Risk & Reliability:** Authored advanced mathematical and graduate-level test cases designed to expose reasoning failure modes in frontier LLMs. Findings directly informed RL reward signals and alignment strategy for next-generation models and produced technical insights for production-readiness risk management.

Research Assistant Professor, Medical Imaging AI

Apr 2019 – May 2022

University of New Mexico – Medical Physics & Radiology | Albuquerque, NM

- **Technical Program Leadership:** Led the AI strategy and computational architecture for a multi-disciplinary center with >\$2M in active grant funding. Governed imaging data pipelines and model development across cardiac, neuro-oncology, and traumatic brain injury studies.
- **Biomedical Data Architecture (Journal Cover Feature):** Rescued a stalled traumatic brain injury study by architecting a custom deep learning framework (U-Net variant) capable of processing noisy, unstandardized biomedical data. Validated the architecture against baselines, earning a cover feature in the Journal of Neurotrauma (2023).
- **Cross-Functional Clinical Integration:** Bridged engineering researchers and clinical stakeholders (Radiology, Neurology, Oncology) to translate AI research into actionable clinical validation studies. Deployed HIPAA-compliant data handling, IRB-governed protocols, and clinical data governance across multi-site imaging studies.

NIH Research Fellow, Cardiovascular MRI

Jul 2016 – Mar 2019

Boston University | Boston, MA

- **Translational Imaging Research:** Developed and validated MRI acquisition and analysis methods for cardiovascular disease, combining physics-informed feature engineering with deep learning, and establishing a framework for choosing deep learning vs. classical methods by efficiency and clinical fit.

SELECTED PROJECTS

- **Browser-Based Edge Inference (napwise.ai).** React + WebLLM application for offline-capable generative AI. Implemented client-side sparse vector search (TF-IDF/cosine similarity) running purely in-browser to rank documents without external APIs.
- **Autonomous Portfolio Manager.** LightGBM proposes tickers/themes; an LLM-orchestrated agent layer reasons over tools (financials, web search, SEC filings) to generate investment rationales; a Python/LLM portfolio manager executes trades via Alpaca with risk guardrails and paper/live modes, plus a Streamlit dashboard for live monitoring and human-in-the-loop trade review. Full CI/CD on GitHub Actions; migrating orchestration to a LangGraph-style graph layer (in progress).
- **Transformer Architecture Series (eriktaylor.github.io).** Author of an open-source engineering series providing reference implementations, mathematical teardowns, and optimized codebases for ViT, CLIP, Diffusion Transformers, DINOv2, and LLM failure mechanisms.

TECHNICAL SKILLS

Agentic AI & LLM Applications: LLM application development (RAG, agents, tool use, function calling) · Claude / Claude Code (daily use) · Anthropic API (via OpenRouter) · LangChain · LangGraph · LlamaIndex · autonomous agent orchestration · Model Context Protocol (MCP) · multi-modal AI · prompt engineering

Model Evaluation & AI Safety: Frontier-model reasoning failure analysis · evaluation harness design · model risk and validation · benchmarking (accuracy / latency / safety / cost trade-offs) · red-teaming · alignment-informed test design

Computer Vision & Edge AI: Detection, depth estimation, segmentation · open-vocabulary detection (OWL-ViT, CLIP) · self-supervised vision (DINO/DINOv2) · YOLO family · on-device / on-sensor inference · split/cascaded inference · quantization (INT8/INT4) · knowledge distillation · ONNX / WASM · browser-based inference (WebLLM, WebGPU) · GPU profiling

Frameworks & Languages: Python (expert) · PyTorch · TensorFlow/Keras · Hugging Face Transformers · scikit-learn · LightGBM/XGBoost · FAISS · JavaScript · SQL · C++ (working) · OpenCV

MLOps & Infrastructure: AWS · GCP · Docker · CI/CD (GitHub Actions) · Cloudflare (Pages, Workers) · distributed training (DDP, FSDP) · experiment tracking (W&B) · Linux/Ubuntu

Domain Knowledge: Medical imaging (MRI, PET, CT) · physics-informed ML · clinical data governance · translational research workflows · edge hardware constraints (memory, latency, power)

STRATEGIC CAPABILITIES

AI Product Strategy & Risk Management: Directing model evaluation and vendor selection · foundation model benchmarking · agentic and algorithmic stress-testing · defining model validation constraints · driving edge vs. cloud deployment strategy.

Cross-Functional Program Leadership: Technical program management and full product lifecycle ownership · stakeholder management across clinical, legal, regulatory, and engineering teams · external partner and vendor coordination · matrixed team leadership.

Clinical AI & Regulatory Awareness: FDA SaMD landscape (PCCP, GMLP, IEC 62304 — actively developing) · EU AI Act high-risk classification · NIST AI RMF · ISO 42001 · HIPAA · IRB-governed research protocols · pursuing IAPP AIGP certification (Q3 2026).

EDUCATION & RESEARCH TRAINING

PhD, Biomedical Engineering — Brown University

Doctoral research in theranostic technologies, integrating advanced medical imaging diagnostics with targeted therapeutics.

NIH Research Fellow, Cardiovascular & Brain MRI — Boston University

Federally funded fellowship developing and evaluating deep learning frameworks for translational medical imaging.

BS, Biomedical Engineering — University of Texas at Austin

Graduate Coursework, Data Science — Harvard University

SELECTED PUBLICATIONS

Selected from 50+ peer-reviewed publications. Full list available on request.

Divani AA, Salazar P, Ikram HA, **Taylor E**, Wilson CM, et al. Non-invasive vagus nerve stimulation improves brain lesion volume and neurobehavioral outcomes in a rat model of traumatic brain injury. *Journal of Neurotrauma*. 2023; 40(13–14): 1481–1494. ★ **Journal Cover Feature**

Taylor EN, Huang N, Lin S, Mortazavi F, Wedeen VJ, et al. Lipid and smooth muscle architectural pathology in the rabbit atherosclerotic vessel wall using Q-space cardiovascular magnetic resonance. *Journal of Cardiovascular Magnetic Resonance*. 2022; 24(1): 74.

Taylor EN, Ding Y, Zhu S, Cheah E, et al. Association between tumor architecture derived from generalized Q-space MRI and survival in glioblastoma. *Oncotarget*. 2017; 8(26): 16296.